1.4.2 ▶ ▶ ▶ ▶ Kinetic Energy of Rigid Bodies (Rotational KE)

▶ The following equation for rotational kinetic energy is required for AP Physics 1, but the derivation is not required. However, nuance from the derivation could be useful.

$$K_{rot} = \frac{1}{2} I \omega^2 \tag{1.55}$$

▶ ▶ ▶ To derive this formula, first consider an arbitrary rotating body (rigid or non-rigid) rotating with speed $\boldsymbol{\omega}$ and translating with speed v_{cm}.

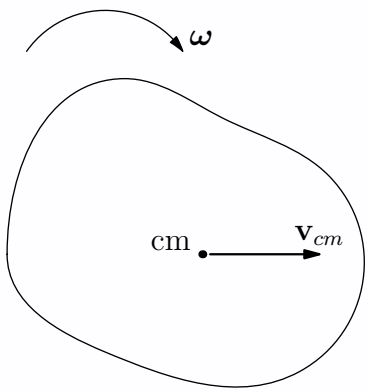

Fig. 1.8

To take the total kinetic energy of this body, we can use the formula derived in 1.3.5:

$$K_{total} = \frac{1}{2} M v_{cm}^2 + \sum \frac{1}{2} m_i v_{icm}^2 \tag{1.56}$$

Remember that this equation is always valid, but it can be difficult to calculate the sum in the second term for every system. So, the objective here is to simplify the sum for certain systems, making it easier to calculate the total energy for these systems. In 1.3.6, we derived the simplified expression for a system of two bodies, and here we derive a simplified expression for rigid bodies.

To begin, we first transform into the reference frame of the center of mass, since we need to find the velocity of the particles relative to the center of mass. Then, we can consider an arbitrary mass element on the body moving at a velocity $\mathbf{u}$ relative to the center of mass at a position $\mathbf{r}_{icm}$ relative to the center of mass. We can split this $\mathbf{u}$ vector into two components: one parallel to $\mathbf{r}_{icm}$ and one perpendicular to $\mathbf{r}_{icm}$. Thus, we have

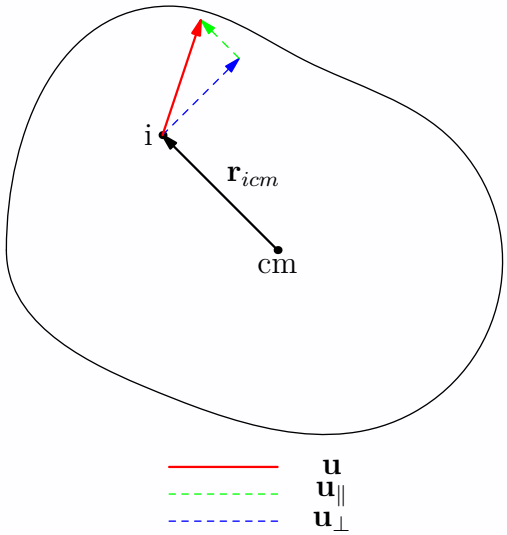

Fig. 1.9

where $\mathbf{u} = \mathbf{u}_{\parallel} + \mathbf{u}_{\perp}$. Since we are using polar coordinates centered at the center of mass, we can represent these vectors as

$$\mathbf{u} = \frac{d\mathbf{r}_{icm}}{dt} + r_{icm}\frac{d\boldsymbol{\theta}}{dt}.$$

Since we are considering a rigd body, $\frac{d\mathbf{r}_{icm}}{dt} = 0$ by definition. Thus, writing our total kinetic energy yields

$$K_{total} = \frac{1}{2}Mv_{cm}^2 + \sum \frac{1}{2}m_i r_{icm}^2 \omega^2.$$

If we define the moment of inertia about the center of mass I_{cm} to be $\sum m_i r_{icm}^2$, we have

$$\boxed{K_{total} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2}. \tag{1.57}$$

This equation states that we can write the total kinetic energy of a rigid body as the kinetic energy due to motion of the center of mass plus the kinetic energy due to rotation about the center of mass.

What if, however, we want to consider the kinetic energy of a rigid body rotating about a fixed pivot? In that case, consider the same rigid body in Fig. 1.8, but this time fixed about point i. Thus, we will also accordingly set our coordinate system to be centered at i.

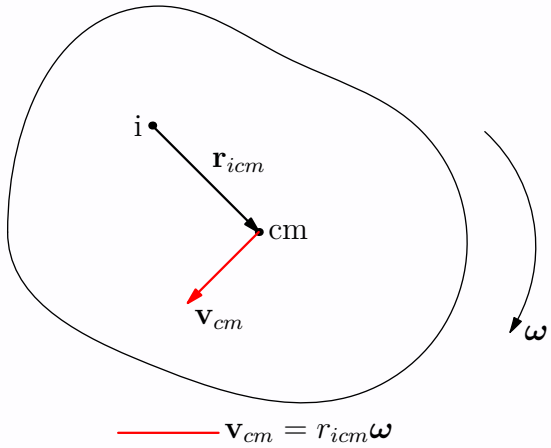

Fig. 1.10

In this case, the center of mass has some velocity $\mathbf{v}_{cm}$ relative to point i. Using the definition of a rigid body and the definition of angular velocity, we have $\mathbf{v}_{cm} = r_{icm}\boldsymbol{\omega}$. Next, since we are solving for rotation about a point separate from the center of mass, it is appropriate to use the Parallel-Axis theorem. Namely, that

$$I_i = I_{cm} + Mr_{icm}^2 \tag{1.58}$$

Plugging I_{cm} into eq. 1.56 gives

$$\begin{aligned} K_{total} &= \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}\omega^2(I_i - Mr_{icm}^2) \\ &= \frac{1}{2}Mv_{cm}^2 - \frac{1}{2}M(r_{icm}\boldsymbol{\omega})^2 + \frac{1}{2}I_i\boldsymbol{\omega}^2 \\ &= \boxed{\frac{1}{2}I_i\boldsymbol{\omega}^2} \end{aligned} \tag{1.59}$$

This expression gives the total kinetic energy of a rigid body rotating about a fixed pivot. Notice that if we transform into the center of mass reference frame here, we essentially have the same scenario as depicted in fig. 1.9.

To summarize, the total kinetic energy of a freely moving rigid body is comprised of two parts: the kinetic energy due to translation of the center of mass $\frac{1}{2}Mv_{cm}^2$ and the kinetic energy due rotation about the center of mass $\frac{1}{2}I_{cm}\omega^2$. For a rigid body rotating about a fixed pivot, the total kinetic energy is simply $\frac{1}{2}I_i\boldsymbol{\omega}^2$ where I_i is the moment of inertia about the fixed pivot ★